



Neural Networks and Deep Learning Introduction

Lecture *Machine Learning* on 10-12.3.2025

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Organization

- Time: 9:00am to 4pm from Mon to Wed
- Lunch break: ~12:00-13:00
- Course material (slides, code, data sets, ...):
 - <https://moodle.uni-greifswald.de/course/view.php?id=5405>
 - <https://bioinf.uni-greifswald.de/bioinf/teaching/NNDL25/KI-Block.tgz>
Extract with `tar xzf KI-Block.tgz`
- Questions and feedback outside of the course (moodle forum):
 - <https://moodle.uni-greifswald.de/mod/forum/view.php?id=146549>
- AppHub Uni Greifswald:
 - <https://apphub.wolke.uni-greifswald.de/>
- Course instructor:
 - Mario Stanke – Professor for Bioinformatics



Prerequisites

- Connection to the network of the University of Greifswald
- Basic programming skills, preferably in Python
- Calculus and linear algebra (derivatives, matrix multiplication)

$$\nabla E(\theta) = \left(\frac{\partial E(\theta)}{\partial \theta_0}, \dots, \frac{\partial E(\theta)}{\partial \theta_n} \right)^T, \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix} \cdot \begin{pmatrix} 3 & 2 & 1 \\ 4 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 11 & 4 & 1 \\ 25 & 10 & 3 \\ 39 & 16 & 5 \end{pmatrix}$$

- Familiarity with `numpy` syntax

```
[1]: import numpy as np

[3]: A = np.array([[1,2],[3,4],[5,6]])
      B = np.array([[3,2,1],[4,1,0]])
      C = np.matmul(A,B)
      C

[3]: array([[11,  4,  1],
           [25, 10,  3],
           [39, 16,  5]])
```



How does image recognition work?

*What are important parameters
I have to choose?*

*What are typical mistakes that
prevents my model from learning?*

How does the model learn?

What are popular machine learning models?

*What is a typical workflow
for a machine learning project?*

Linear Regression

Stochastic Gradient Descend

Neural Networks

Convolutional Neural Networks

Image Derivative

Transformers and Attention

What does the machine learn?
*What are useful applications
for machine learning?*

*How do I customize a machine learning
model for my purposes?*

*How can I implement a machine learning
workflow with a high level API?*

How do I prevent overfitting?

What is deep learning?

How can I fool a trained model?